# Personal Statement

I am Joes Francy Reeba Sahayam, a third-year undergraduate student studying Computer Science and Engineering with a focus on Artificial Intelligence and Machine Learning at Joy University, Tirunelveli. I have an SGPA of 9.54. Since the beginning of my course, I have been eager to understand how computational systems learn and interact with data in real life. This curiosity has led me to develop a habit of learning through research and self-exploration.

When I started my undergraduate studies, I was not very clear about which research area I wanted to focus on. I spent time exploring different topics by reading papers, trying small projects, and learning how researchers think about problems and improve their ideas over time. Although this phase was sometimes challenging, it helped me understand how research works and gradually shaped my interests. Through self-learning and hands-on experimentation, I built a strong foundation in machine learning and realised that I am particularly interested in computer vision, reinforcement learning, and generative models, where theory and practical experimentation come together. To strengthen my fundamentals, I learned linear algebra for machine learning through courses by Andrew Ng, which helped me better understand the mathematics behind these approaches.

One experience that had a big impact on me was working with Reinforcement Learning with Human Feedback (RLHF). Understanding how rewards and KL-based penalties are adjusted during training helped me appreciate the delicate balance between improving a model and keeping it stable and aligned. Around the same time, I explored diffusion-based text-to-image models and even tried to implement a simple version of Stable Diffusion, which gave me hands-on insight into generative modelling and iterative experimentation.

Along with my interest in research, I worked on applied projects that helped me see how theoretical ideas can solve real-world problems. I developed a Heart Disease Prediction System using machine learning. I tested different algorithms and found that Random Forest produced the most reliable results. I also created a Java-based Stampede Prevention System. This project was inspired by the tragic stampede incidents in our country at that time, which led to many deaths. My goal was to improve crowd safety by using alerts based on population density and noise for early warning. These projects taught me to think beyond just accuracy and performance. I started to focus on system behaviour, robustness, and the limitations of the real world. I have strengthened my coding skills using Python, Java, and Jupyter Notebook, along with libraries like NumPy, Pandas, and scikit-learn, implementing ML models and small projects. Coding has taught me patience, logical thinking, and how to turn ideas into working solutions. Coming from a university with limited exposure to large research laboratories, I relied heavily on self-learning, curiosity, and discipline to gradually build my research skills. To bridge this gap, I took part in workshops and programs from top institutions IITs, IIITs, IISc, NITs, and VIT, and presented my work at national competitions, where I won first prizes for paper and poster presentations. These experiences gave me confidence and strengthened my dedication to research.

I am applying for the NUS internship under the IRIS program because of its strong research culture, academic rigor, and global reputation. I have always aspired to study at NUS, and the IRIS internship provides an opportunity to experience its research environment firsthand and learn from experienced researchers.

If selected, I will contribute my full effort through careful experimentation, thoughtful implementation, and consistent dedication. I am excited to learn, take guidance, and grow as a researcher. This internship would be an important step toward my long-term goal of pursuing research-oriented higher studies, including a potential PhD, and I would be honoured to contribute meaningfully to the ongoing research at NUS.